

Economic Value of Adaptivity

Research Overview

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THE BUSINESS PROBLEM

Retail category managers make allocation decisions under conditions that no static model fully captures. Demand for a product like eggs looks nothing like demand for a seasonal product like fresh mangoes, which looks nothing like demand for a consumer electronics item entering a new product lifecycle phase. Replenishment cycles, promotional windows, supplier constraints, and competitive pricing all interact with demand in ways that shift over time. The category manager sitting above all of this must decide how much inventory to allocate, when to allocate it, and how to respond when demand signals deviate from prior expectations.

Most deployed retail allocation systems handle this with fixed rules or one-shot optimization models calibrated on historical data. These systems are not designed to update as demand evolves within a planning horizon or to learn across planning horizons as structural demand patterns shift. The gap they leave is not a forecasting gap. It is a decision gap, more specifically the difference between what an adaptive system could deliver if it were allowed to update its policy as uncertainty resolves, and what a static system actually delivers when it cannot.

That gap has real commercial consequences. In category management, over-allocation ties up working capital and drives markdowns. Under-allocation drives stockouts and lost revenue. Promotional timing decisions made on stale demand assumptions generate spend without verifiable lift. The cost of getting allocation wrong is not abstract. It is measurable in margin, in stockout rate, and in promotional ROI.

THE RESEARCH QUESTION

The Economic Value of Adaptivity (EVA) is a formal construct that quantifies the value of adaptive decision-making in retail allocation. EVA measures the gap between the theoretical maximum value delivered by a fully adaptive allocation policy and the value delivered by a static policy calibrated on prior demand beliefs.

EVA decomposes into two components. The first captures the value of reacting to demand signals as they arrive within a planning horizon. The second captures the additional value of anticipating structural demand shifts across planning horizons. These two components are not equivalent and do not always co-occur. Understanding which component dominates in each demand environment is what determines which policy design is appropriate.

The research also addresses a second question that the allocation literature has largely left open: how much of the theoretical EVA is realized when a human decision maker acts on algorithmic recommendations rather than implementing the policy directly? The representation choices that govern how recommendations are communicated includes what information is shown, in what form, at what level of uncertainty and determine the fraction of theoretical value that reaches a business decision. That fraction is itself a measurable and designable quantity.

THE RESEARCH ARCHITECTURE

SKUFlux is the four-layer system operationalizing this research.

The demand foundation derives store-level demand from consumer behavioural principles, grounding the simulation environment in theory rather than reduced-form assumption. The derivation connects individual purchasing decisions to aggregate demand dynamics in a way that preserves the interpretation of key parameters as behaviourally meaningful quantities such as habit formation and preference persistence.

The causal layer identifies the structural drivers of allocation decisions. Demand-side and supply-side analysis are treated as methodologically distinct tracks. The supply-side operates under an inventory constraint that transforms the problem from a replenishment decision into a genuine allocation decision across a planning horizon.

The optimization layer evaluates allocation policies across four demand environments, each designed to stress a different aspect of adaptive decision-making. The first is a stable high-variance

environment where within-horizon reactivity dominates (eggs are the commercial analogue). The second is a seasonal environment with an outdated prior where cross-horizon anticipation dominates (fresh mangoes). The third is a regime-switching environment requiring state detection (a consumer electronics product in transition). The fourth combines structural breaks with persistent uncertainty - product experiencing a lasting demand shift. Three policies spanning the adaptivity spectrum are evaluated against each environment. Real-world commercial deployments anchor each end of that spectrum.

The human interface layer addresses how recommendation representation affects the realized fraction of theoretical EVA. Two completed empirical studies under Professor Jana Schaich Borg at Duke have established that representation choices systematically affect the quality of human decisions in measurable ways. Descriptive information framing creates overconfidence in incorrect conclusions. Take-home framing increases correct conclusion rates. These findings directly inform interface design for the adaptive allocation system.

CURRENT RESEARCH ACTIVITY

The demand simulation environment is under active development. Optimization formulation depth is grounded in two active engagements: the MOPTA 2026 competition at Lehigh University, applying constrained network optimization to a fleet routing problem, and the Durham Public Schools facility location project, a multi-objective capacitated facility location model with a Pareto frontier across equity, stability, and distance objectives. Both demonstrate constraint formulation capability at the complexity level the allocation optimization layer requires.

The Bass Connections empirical program is ongoing with two studies complete and a paper in progress under Professor Jana Schaich Borg. A follow-up eye-tracking study validating Phase 1 findings through attentional mechanism analysis is designed and ready to execute.

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Full research documentation and simulation design available on request.